# MSG No. 226.123 $Fm\bar{3}c1'$ [ Type II, cubic ]

Table 1: Wyckoff site: 8a, site symmetry: 4321'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,12,14,16,17,18,53,54,56,61,70,71,98,103,105,111,115,120,147,148,154,155,164,165, |
| | | 193,204,206,208,209,210,245,246,248,253,262,263,290,295,297,303,307,312,339,340,346,347,356,357] |
| 2 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [2,7,9,15,19,24,51,52,58,59,68,69,97,108,110,112,113,114,149,150,152,157,166,167, |
| | | 194,199,201,207,211,216,243,244,250,251,260,261,289,300,302,304,305,306,341,342,344,349,358,359] |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [3,4,10,11,20,21,50,55,57,63,67,72,101,102,104,109,118,119,145,156,158,160,161,162, |
| | | 195,196,202,203,212,213,242,247,249,255,259,264,293,294,296,301,310,311,337,348,350,352,353,354] |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [5,6,8,13,22,23,49,60,62,64,65,66,99,100,106,107,116,117,146,151,153,159,163,168, |
| | | 197,198,200,205,214,215,241,252,254,256,257,258,291,292,298,299,308,309,338,343,345,351,355,360] |
| 5 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [25,36,38,40,41,42,77,78,80,85,94,95,122,127,129,135,139,144,171,172,178,179,188,189, |
| | | 217,228,230,232,233,234,269,270,272,277,286,287,314,319,321,327,331,336,363,364,370,371,380,381] |
| 6 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [26,31,33,39,43,48,75,76,82,83,92,93,121,132,134,136,137,138,173,174,176,181,190,191, |
| | | 218,223,225,231,235,240,267,268,274,275,284,285,313,324,326,328,329,330,365,366,368,373,382,383] |
| 7 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [27,28,34,35,44,45,74,79,81,87,91,96,125,126,128,133,142,143,169,180,182,184,185,186, |
| | | 219,220,226,227,236,237,266,271,273,279,283,288,317,318,320,325,334,335,361,372,374,376,377,378] |
| 8 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [29,30,32,37,46,47,73,84,86,88,89,90,123,124,130,131,140,141,170,175,177,183,187,192, |
| | | 221,222,224,229,238,239,265,276,278,280,281,282,315,316,322,323,332,333,362,367,369,375,379,384] |

Table 2: Wyckoff site: 8b, site symmetry: m-3.1'

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,8,9,10,17,18,19,20,21,22,23,24,25,32,33,34,41,42,43,44,45,46,47,48, |
| | | 193,200,201,202,209,210,211,212,213,214,215,216,217,224,225,226,233,234,235,236,237,238,239,240] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [2,3,4,5,6,7,11,12,13,14,15,16,26,27,28,29,30,31,35,36,37,38,39,40, |
| | | 194,195,196,197,198,199,203,204,205,206,207,208,218,219,220,221,222,223,227,228,229,230,231,232] |
| 3 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [49,56,57,58,65,66,67,68,69,70,71,72,73,80,81,82,89,90,91,92,93,94,95,96, |
| | | 241,248,249,250,257,258,259,260,261,262,263,264,265,272,273,274,281,282,283,284,285,286,287,288] |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | [50,51,52,53,54,55,59,60,61,62,63,64,74,75,76,77,78,79,83,84,85,86,87,88, |
| | | 242,243,244,245,246,247,251,252,253,254,255,256,266,267,268,269,270,271,275,276,277,278,279,280] |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [97,104,105,106,113,114,115,116,117,118,119,120,121,128,129,130,137,138,139,140,141,142,143,144, |
| | | 289,296,297,298,305,306,307,308,309,310,311,312,313,320,321,322,329,330,331,332,333,334,335,336] |
| 6 | $\left[0, \frac{1}{2}, 0\right]$ | [98,99,100,101,102,103,107,108,109,110,111,112,122,123,124,125,126,127,131,132,133,134,135,136, |
| | | 290,291,292,293,294,295,299,300,301,302,303,304,314,315,316,317,318,319,323,324,325,326,327,328] |
| 7 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [145,152,153,154,161,162,163,164,165,166,167,168,169,176,177,178,185,186,187,188,189,190,191,192, |
| | | 337,344,345,346,353,354,355,356,357,358,359,360,361,368,369,370,377,378,379,380,381,382,383,384] |
| 8 | $\left[0, 0, \frac{1}{2}\right]$ | [146,147,148,149,150,151,155,156,157,158,159,160,170,171,172,173,174,175,179,180,181,182,183,184, |
| | | 338,339,340,341,342,343,347,348,349,350,351,352,362,363,364,365,366,367,371,372,373,374,375,376] |

Table 3: Wyckoff site: 24c, site symmetry: -4m.21'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, 0\right]$ | [1,8,33,34,61,62,74,75,193,200,225,226,253,254,266,267] |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$ | [2,3,37,38,57,58,73,80,194,195,229,230,249,250,265,272] |

*continued ...*

Table 3

| No. | position | mapping |
|---|---|---|
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$ | [4,16,29,39,162,165,187,191,196,208,221,231,354,357,379,383] |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$ | [5,15,28,40,163,167,186,189,197,207,220,232,355,359,378,381] |
| 5 | $\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$ | [6,11,31,36,116,118,137,144,198,203,223,228,308,310,329,336] |
| 6 | $\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$ | [7,12,30,35,113,120,140,142,199,204,222,227,305,312,332,334] |
| 7 | $\left[\frac{3}{4}, 0, 0\right]$ | [9,10,25,32,50,51,85,86,201,202,217,224,242,243,277,278] |
| 8 | $\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$ | [13,14,26,27,49,56,81,82,205,206,218,219,241,248,273,274] |
| 9 | $\left[0, \frac{1}{4}, 0\right]$ | [17,24,44,46,103,108,126,131,209,216,236,238,295,300,318,323] |
| 10 | $\left[0, 0, \frac{1}{4}\right]$ | [18,21,43,47,148,160,173,183,210,213,235,239,340,352,365,375] |
| 11 | $\left[0, 0, \frac{3}{4}\right]$ | [19,23,42,45,149,159,172,184,211,215,234,237,341,351,364,376] |
| 12 | $\left[0, \frac{3}{4}, 0\right]$ | [20,22,41,48,102,107,127,132,212,214,233,240,294,299,319,324] |
| 13 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [52,64,77,87,114,117,139,143,244,256,269,279,306,309,331,335] |
| 14 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [53,63,76,88,115,119,138,141,245,255,268,280,307,311,330,333] |
| 15 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [54,59,79,84,164,166,185,192,246,251,271,276,356,358,377,384] |
| 16 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | [55,60,78,83,161,168,188,190,247,252,270,275,353,360,380,382] |
| 17 | $\left[0, \frac{3}{4}, \frac{1}{2}\right]$ | [65,72,92,94,151,156,174,179,257,264,284,286,343,348,366,371] |
| 18 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [66,69,91,95,100,112,125,135,258,261,283,287,292,304,317,327] |
| 19 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [67,71,90,93,101,111,124,136,259,263,282,285,293,303,316,328] |
| 20 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [68,70,89,96,150,155,175,180,260,262,281,288,342,347,367,372] |
| 21 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [97,104,129,130,157,158,170,171,289,296,321,322,349,350,362,363] |
| 22 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [98,99,133,134,153,154,169,176,290,291,325,326,345,346,361,368] |
| 23 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [105,106,121,128,146,147,181,182,297,298,313,320,338,339,373,374] |
| 24 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [109,110,122,123,145,152,177,178,301,302,314,315,337,344,369,370] |

Table 4: Wyckoff site: 24d, site symmetry: 4/m..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | [1,32,56,73,98,123,147,170,193,224,248,265,290,315,339,362] |
| 2 | $\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$ | [2,27,51,74,97,128,152,169,194,219,243,266,289,320,344,361] |
| 3 | $\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$ | [3,26,50,75,104,121,145,176,195,218,242,267,296,313,337,368] |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | [4,40,67,95,119,139,160,172,196,232,259,287,311,331,352,364] |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | [5,39,66,93,117,138,159,173,197,231,258,285,309,330,351,365] |
| 6 | $\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$ | [6,35,65,96,107,126,168,185,198,227,257,288,299,318,360,377] |
| 7 | $\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$ | [7,36,68,94,108,127,166,188,199,228,260,286,300,319,358,380] |
| 8 | $\left[0, \frac{3}{4}, \frac{3}{4}\right]$ | [8,25,49,80,99,122,146,171,200,217,241,272,291,314,338,363] |
| 9 | $\left[0, \frac{1}{4}, \frac{3}{4}\right]$ | [9,34,58,81,110,133,157,182,201,226,250,273,302,325,349,374] |
| 10 | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | [10,33,57,82,109,134,158,181,202,225,249,274,301,326,350,373] |
| 11 | $\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$ | [11,30,72,89,102,131,161,192,203,222,264,281,294,323,353,384] |
| 12 | $\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$ | [12,31,70,92,103,132,164,190,204,223,262,284,295,324,356,382] |
| 13 | $\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$ | [13,38,62,85,106,129,153,178,205,230,254,277,298,321,345,370] |
| 14 | $\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$ | [14,37,61,86,105,130,154,177,206,229,253,278,297,322,346,369] |
| 15 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | [15,29,69,90,114,141,149,183,207,221,261,282,306,333,341,375] |
| 16 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | [16,28,71,91,115,143,148,184,208,220,263,283,307,335,340,376] |
| 17 | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | [17,48,54,83,120,137,155,174,209,240,246,275,312,329,347,366] |
| 18 | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | [18,45,53,87,111,125,165,186,210,237,245,279,303,317,357,378] |
| 19 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | [19,47,52,88,112,124,167,187,211,239,244,280,304,316,359,379] |

*continued ...*

Table 4

| No. | position | mapping |
|---|---|---|
| 20 | $\left[\frac{3}{4}, 0, \frac{1}{4}\right]$ | [20,46,55,84,118,140,156,175,212,238,247,276,310,332,348,367] |
| 21 | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | [21,42,63,77,101,135,162,189,213,234,255,269,293,327,354,381] |
| 22 | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | [22,44,60,79,116,142,151,180,214,236,252,271,308,334,343,372] |
| 23 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | [23,43,64,76,100,136,163,191,215,235,256,268,292,328,355,383] |
| 24 | $\left[\frac{3}{4}, 0, \frac{3}{4}\right]$ | [24,41,59,78,113,144,150,179,216,233,251,270,305,336,342,371] |

Table 5: Wyckoff site: 48e, site symmetry: 2mm..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | [1,8,33,34,193,200,225,226] |
| 2 | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [2,3,37,38,194,195,229,230] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | [4,16,29,39,196,208,221,231] |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | [5,15,28,40,197,207,220,232] |
| 5 | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | [6,11,31,36,198,203,223,228] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | [7,12,30,35,199,204,222,227] |
| 7 | $[-x, 0, 0]$ | [9,10,25,32,201,202,217,224] |
| 8 | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | [13,14,26,27,205,206,218,219] |
| 9 | $[0, x, 0]$ | [17,24,44,46,209,216,236,238] |
| 10 | $[0, 0, x]$ | [18,21,43,47,210,213,235,239] |
| 11 | $[0, 0, -x]$ | [19,23,42,45,211,215,234,237] |
| 12 | $[0, -x, 0]$ | [20,22,41,48,212,214,233,240] |
| 13 | $\left[x, \frac{1}{2}, \frac{1}{2}\right]$ | [49,56,81,82,241,248,273,274] |
| 14 | $\left[x+\frac{1}{2}, 0, 0\right]$ | [50,51,85,86,242,243,277,278] |
| 15 | $\left[\frac{1}{2}, 0, -x\right]$ | [52,64,77,87,244,256,269,279] |
| 16 | $\left[\frac{1}{2}, 0, x\right]$ | [53,63,76,88,245,255,268,280] |
| 17 | $\left[\frac{1}{2}, x, 0\right]$ | [54,59,79,84,246,251,271,276] |
| 18 | $\left[\frac{1}{2}, -x, 0\right]$ | [55,60,78,83,247,252,270,275] |
| 19 | $\left[-x, \frac{1}{2}, \frac{1}{2}\right]$ | [57,58,73,80,249,250,265,272] |
| 20 | $\left[\frac{1}{2}-x, 0, 0\right]$ | [61,62,74,75,253,254,266,267] |
| 21 | $\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$ | [65,72,92,94,257,264,284,286] |
| 22 | $\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$ | [66,69,91,95,258,261,283,287] |
| 23 | $\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$ | [67,71,90,93,259,263,282,285] |
| 24 | $\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$ | [68,70,89,96,260,262,281,288] |
| 25 | $\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$ | [97,104,129,130,289,296,321,322] |
| 26 | $\left[x, \frac{1}{2}, 0\right]$ | [98,99,133,134,290,291,325,326] |
| 27 | $\left[0, \frac{1}{2}, -x\right]$ | [100,112,125,135,292,304,317,327] |
| 28 | $\left[0, \frac{1}{2}, x\right]$ | [101,111,124,136,293,303,316,328] |
| 29 | $\left[0, x+\frac{1}{2}, 0\right]$ | [102,107,127,132,294,299,319,324] |
| 30 | $\left[0, \frac{1}{2}-x, 0\right]$ | [103,108,126,131,295,300,318,323] |
| 31 | $\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$ | [105,106,121,128,297,298,313,320] |
| 32 | $\left[-x, \frac{1}{2}, 0\right]$ | [109,110,122,123,301,302,314,315] |
| 33 | $\left[\frac{1}{2}, x, \frac{1}{2}\right]$ | [113,120,140,142,305,312,332,334] |
| 34 | $\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$ | [114,117,139,143,306,309,331,335] |
| 35 | $\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$ | [115,119,138,141,307,311,330,333] |
| 36 | $\left[\frac{1}{2}, -x, \frac{1}{2}\right]$ | [116,118,137,144,308,310,329,336] |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 37 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | [145,152,177,178,337,344,369,370] |
| 38 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | [146,147,181,182,338,339,373,374] |
| 39 | $\left[0,\ 0,\ \frac{1}{2}-x\right]$ | [148,160,173,183,340,352,365,375] |
| 40 | $\left[0,\ 0,\ x+\frac{1}{2}\right]$ | [149,159,172,184,341,351,364,376] |
| 41 | $\left[0,\ x,\ \frac{1}{2}\right]$ | [150,155,175,180,342,347,367,372] |
| 42 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | [151,156,174,179,343,348,366,371] |
| 43 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | [153,154,169,176,345,346,361,368] |
| 44 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | [157,158,170,171,349,350,362,363] |
| 45 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$ | [161,168,188,190,353,360,380,382] |
| 46 | $\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$ | [162,165,187,191,354,357,379,383] |
| 47 | $\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$ | [163,167,186,189,355,359,378,381] |
| 48 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | [164,166,185,192,356,358,377,384] |

Table 6: Wyckoff site: 48f, site symmetry: 4..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | [1,56,98,147,193,248,290,339] |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$ | [2,51,97,152,194,243,289,344] |
| 3 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$ | [3,50,104,145,195,242,296,337] |
| 4 | $\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$ | [4,67,119,160,196,259,311,352] |
| 5 | $\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$ | [5,66,117,159,197,258,309,351] |
| 6 | $\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [6,65,107,168,198,257,299,360] |
| 7 | $\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [7,68,108,166,199,260,300,358] |
| 8 | $\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | [8,49,99,146,200,241,291,338] |
| 9 | $\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$ | [9,58,110,157,201,250,302,349] |
| 10 | $\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | [10,57,109,158,202,249,301,350] |
| 11 | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | [11,72,102,161,203,264,294,353] |
| 12 | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [12,70,103,164,204,262,295,356] |
| 13 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | [13,62,106,153,205,254,298,345] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | [14,61,105,154,206,253,297,346] |
| 15 | $\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$ | [15,69,114,149,207,261,306,341] |
| 16 | $\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$ | [16,71,115,148,208,263,307,340] |
| 17 | $\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$ | [17,54,120,155,209,246,312,347] |
| 18 | $\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$ | [18,53,111,165,210,245,303,357] |
| 19 | $\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$ | [19,52,112,167,211,244,304,359] |
| 20 | $\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$ | [20,55,118,156,212,247,310,348] |
| 21 | $\left[\frac{3}{4},\ \frac{3}{4},\ x\right]$ | [21,63,101,162,213,255,293,354] |
| 22 | $\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$ | [22,60,116,151,214,252,308,343] |
| 23 | $\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$ | [23,64,100,163,215,256,292,355] |
| 24 | $\left[\frac{3}{4},\ x,\ \frac{3}{4}\right]$ | [24,59,113,150,216,251,305,342] |
| 25 | $\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | [25,80,122,171,217,272,314,363] |
| 26 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | [26,75,121,176,218,267,313,368] |
| 27 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$ | [27,74,128,169,219,266,320,361] |
| 28 | $\left[\frac{1}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$ | [28,91,143,184,220,283,335,376] |
| 29 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$ | [29,90,141,183,221,282,333,375] |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 30 | $\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [30,89,131,192,222,281,323,384] |
| 31 | $\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | [31,92,132,190,223,284,324,382] |
| 32 | $\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | [32,73,123,170,224,265,315,362] |
| 33 | $\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | [33,82,134,181,225,274,326,373] |
| 34 | $\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$ | [34,81,133,182,226,273,325,374] |
| 35 | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [35,96,126,185,227,288,318,377] |
| 36 | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [36,94,127,188,228,286,319,380] |
| 37 | $\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$ | [37,86,130,177,229,278,322,369] |
| 38 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$ | [38,85,129,178,230,277,321,370] |
| 39 | $\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$ | [39,93,138,173,231,285,330,365] |
| 40 | $\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$ | [40,95,139,172,232,287,331,364] |
| 41 | $\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$ | [41,78,144,179,233,270,336,371] |
| 42 | $\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$ | [42,77,135,189,234,269,327,381] |
| 43 | $\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$ | [43,76,136,191,235,268,328,383] |
| 44 | $\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$ | [44,79,142,180,236,271,334,372] |
| 45 | $\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$ | [45,87,125,186,237,279,317,378] |
| 46 | $\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$ | [46,84,140,175,238,276,332,367] |
| 47 | $\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$ | [47,88,124,187,239,280,316,379] |
| 48 | $\left[\frac{1}{4},\ -x,\ \frac{1}{4}\right]$ | [48,83,137,174,240,275,329,366] |

Table 7: Wyckoff site: 64g, site symmetry: .3.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ x]$ | [1,17,18,193,209,210] |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [2,7,15,194,199,207] |
| 3 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [3,4,11,195,196,203] |
| 4 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [5,6,13,197,198,205] |
| 5 | $[x,\ -x,\ -x]$ | [8,22,23,200,214,215] |
| 6 | $[-x,\ x,\ -x]$ | [9,19,24,201,211,216] |
| 7 | $[-x,\ -x,\ x]$ | [10,20,21,202,212,213] |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [12,14,16,204,206,208] |
| 9 | $[-x,\ -x,\ -x]$ | [25,41,42,217,233,234] |
| 10 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [26,31,39,218,223,231] |
| 11 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [27,28,35,219,220,227] |
| 12 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [29,30,37,221,222,229] |
| 13 | $[-x,\ x,\ x]$ | [32,46,47,224,238,239] |
| 14 | $[x,\ -x,\ x]$ | [33,43,48,225,235,240] |
| 15 | $[x,\ x,\ -x]$ | [34,44,45,226,236,237] |
| 16 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [36,38,40,228,230,232] |
| 17 | $\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [49,65,66,241,257,258] |
| 18 | $\left[x+\frac{1}{2},\ -x,\ x\right]$ | [50,55,63,242,247,255] |
| 19 | $\left[x+\frac{1}{2},\ x,\ -x\right]$ | [51,52,59,243,244,251] |
| 20 | $\left[\frac{1}{2}-x,\ x,\ x\right]$ | [53,54,61,245,246,253] |
| 21 | $\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [56,70,71,248,262,263] |
| 22 | $\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [57,67,72,249,259,264] |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 23 | $\left[-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [58,68,69,250,260,261] |
| 24 | $\left[\frac{1}{2}-x,\ -x,\ -x\right]$ | [60,62,64,252,254,256] |
| 25 | $\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [73,89,90,265,281,282] |
| 26 | $\left[\frac{1}{2}-x,\ x,\ -x\right]$ | [74,79,87,266,271,279] |
| 27 | $\left[\frac{1}{2}-x,\ -x,\ x\right]$ | [75,76,83,267,268,275] |
| 28 | $\left[x+\frac{1}{2},\ -x,\ -x\right]$ | [77,78,85,269,270,277] |
| 29 | $\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [80,94,95,272,286,287] |
| 30 | $\left[x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [81,91,96,273,283,288] |
| 31 | $\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [82,92,93,274,284,285] |
| 32 | $\left[x+\frac{1}{2},\ x,\ x\right]$ | [84,86,88,276,278,280] |
| 33 | $\left[x+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$ | [97,113,114,289,305,306] |
| 34 | $\left[x,\ \frac{1}{2}-x,\ x\right]$ | [98,103,111,290,295,303] |
| 35 | $\left[x,\ x+\frac{1}{2},\ -x\right]$ | [99,100,107,291,292,299] |
| 36 | $\left[-x,\ x+\frac{1}{2},\ x\right]$ | [101,102,109,293,294,301] |
| 37 | $\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-x\right]$ | [104,118,119,296,310,311] |
| 38 | $\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$ | [105,115,120,297,307,312] |
| 39 | $\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$ | [106,116,117,298,308,309] |
| 40 | $\left[-x,\ \frac{1}{2}-x,\ -x\right]$ | [108,110,112,300,302,304] |
| 41 | $\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-x\right]$ | [121,137,138,313,329,330] |
| 42 | $\left[-x,\ x+\frac{1}{2},\ -x\right]$ | [122,127,135,314,319,327] |
| 43 | $\left[-x,\ \frac{1}{2}-x,\ x\right]$ | [123,124,131,315,316,323] |
| 44 | $\left[x,\ \frac{1}{2}-x,\ -x\right]$ | [125,126,133,317,318,325] |
| 45 | $\left[\frac{1}{2}-x,\ x,\ x+\frac{1}{2}\right]$ | [128,142,143,320,334,335] |
| 46 | $\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$ | [129,139,144,321,331,336] |
| 47 | $\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-x\right]$ | [130,140,141,322,332,333] |
| 48 | $\left[x,\ x+\frac{1}{2},\ x\right]$ | [132,134,136,324,326,328] |
| 49 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$ | [145,161,162,337,353,354] |
| 50 | $\left[x,\ -x,\ x+\frac{1}{2}\right]$ | [146,151,159,338,343,351] |
| 51 | $\left[x,\ x,\ \frac{1}{2}-x\right]$ | [147,148,155,339,340,347] |
| 52 | $\left[-x,\ x,\ x+\frac{1}{2}\right]$ | [149,150,157,341,342,349] |
| 53 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$ | [152,166,167,344,358,359] |
| 54 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$ | [153,163,168,345,355,360] |
| 55 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$ | [154,164,165,346,356,357] |
| 56 | $\left[-x,\ -x,\ \frac{1}{2}-x\right]$ | [156,158,160,348,350,352] |
| 57 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -x\right]$ | [169,185,186,361,377,378] |
| 58 | $\left[-x,\ x,\ \frac{1}{2}-x\right]$ | [170,175,183,362,367,375] |
| 59 | $\left[-x,\ -x,\ x+\frac{1}{2}\right]$ | [171,172,179,363,364,371] |
| 60 | $\left[x,\ -x,\ \frac{1}{2}-x\right]$ | [173,174,181,365,366,373] |
| 61 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x\right]$ | [176,190,191,368,382,383] |
| 62 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x\right]$ | [177,187,192,369,379,384] |
| 63 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$ | [178,188,189,370,380,381] |
| 64 | $\left[x,\ x,\ x+\frac{1}{2}\right]$ | [180,182,184,372,374,376] |

Table 8: Wyckoff site: `96h`, site symmetry: `..21'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, y, y\right]$ | `[1,61,193,253]` |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}-y, y+\frac{1}{2}\right]$ | `[2,58,194,250]` |
| 3 | $\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{2}-y\right]$ | `[3,57,195,249]` |
| 4 | $\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$ | `[4,162,196,354]` |
| 5 | $\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{3}{4}\right]$ | `[5,163,197,355]` |
| 6 | $\left[\frac{1}{2}-y, \frac{3}{4}, y+\frac{1}{2}\right]$ | `[6,116,198,308]` |
| 7 | $\left[y+\frac{1}{2}, \frac{1}{4}, y+\frac{1}{2}\right]$ | `[7,113,199,305]` |
| 8 | $\left[\frac{1}{4}, -y, -y\right]$ | `[8,62,200,254]` |
| 9 | $\left[\frac{3}{4}, y, -y\right]$ | `[9,51,201,243]` |
| 10 | $\left[\frac{3}{4}, -y, y\right]$ | `[10,50,202,242]` |
| 11 | $\left[y+\frac{1}{2}, \frac{3}{4}, \frac{1}{2}-y\right]$ | `[11,118,203,310]` |
| 12 | $\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$ | `[12,120,204,312]` |
| 13 | $\left[\frac{1}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$ | `[13,49,205,241]` |
| 14 | $\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$ | `[14,56,206,248]` |
| 15 | $\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$ | `[15,167,207,359]` |
| 16 | $\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$ | `[16,165,208,357]` |
| 17 | $\left[y, \frac{1}{4}, y\right]$ | `[17,103,209,295]` |
| 18 | $\left[y, y, \frac{1}{4}\right]$ | `[18,148,210,340]` |
| 19 | $\left[-y, y, \frac{3}{4}\right]$ | `[19,149,211,341]` |
| 20 | $\left[-y, \frac{3}{4}, y\right]$ | `[20,102,212,294]` |
| 21 | $\left[-y, -y, \frac{1}{4}\right]$ | `[21,160,213,352]` |
| 22 | $\left[y, \frac{3}{4}, -y\right]$ | `[22,107,214,299]` |
| 23 | $\left[y, -y, \frac{3}{4}\right]$ | `[23,159,215,351]` |
| 24 | $\left[-y, \frac{1}{4}, -y\right]$ | `[24,108,216,300]` |
| 25 | $\left[\frac{3}{4}, -y, -y\right]$ | `[25,85,217,277]` |
| 26 | $\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{2}-y\right]$ | `[26,82,218,274]` |
| 27 | $\left[\frac{1}{4}, \frac{1}{2}-y, y+\frac{1}{2}\right]$ | `[27,81,219,273]` |
| 28 | $\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{3}{4}\right]$ | `[28,186,220,378]` |
| 29 | $\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{4}\right]$ | `[29,187,221,379]` |
| 30 | $\left[y+\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-y\right]$ | `[30,140,222,332]` |
| 31 | $\left[\frac{1}{2}-y, \frac{3}{4}, \frac{1}{2}-y\right]$ | `[31,137,223,329]` |
| 32 | $\left[\frac{3}{4}, y, y\right]$ | `[32,86,224,278]` |
| 33 | $\left[\frac{1}{4}, -y, y\right]$ | `[33,75,225,267]` |
| 34 | $\left[\frac{1}{4}, y, -y\right]$ | `[34,74,226,266]` |
| 35 | $\left[\frac{1}{2}-y, \frac{1}{4}, y+\frac{1}{2}\right]$ | `[35,142,227,334]` |
| 36 | $\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$ | `[36,144,228,336]` |
| 37 | $\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$ | `[37,73,229,265]` |
| 38 | $\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$ | `[38,80,230,272]` |
| 39 | $\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{4}\right]$ | `[39,191,231,383]` |
| 40 | $\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$ | `[40,189,232,381]` |
| 41 | $\left[-y, \frac{3}{4}, -y\right]$ | `[41,127,233,319]` |
| 42 | $\left[-y, -y, \frac{3}{4}\right]$ | `[42,172,234,364]` |
| 43 | $\left[y, -y, \frac{1}{4}\right]$ | `[43,173,235,365]` |
| 44 | $\left[y, \frac{1}{4}, -y\right]$ | `[44,126,236,318]` |
| 45 | $\left[y, y, \frac{3}{4}\right]$ | `[45,184,237,376]` |
| 46 | $\left[-y, \frac{1}{4}, y\right]$ | `[46,131,238,323]` |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 47 | $\left[-y,\, y,\, \frac{1}{4}\right]$ | [47,183,239,375] |
| 48 | $\left[y,\, \frac{3}{4},\, y\right]$ | [48,132,240,324] |
| 49 | $\left[y+\frac{1}{2},\, y,\, \frac{3}{4}\right]$ | [52,114,244,306] |
| 50 | $\left[\frac{1}{2}-y,\, y,\, \frac{1}{4}\right]$ | [53,115,245,307] |
| 51 | $\left[\frac{1}{2}-y,\, \frac{1}{4},\, y\right]$ | [54,164,246,356] |
| 52 | $\left[y+\frac{1}{2},\, \frac{3}{4},\, y\right]$ | [55,161,247,353] |
| 53 | $\left[y+\frac{1}{2},\, \frac{1}{4},\, -y\right]$ | [59,166,251,358] |
| 54 | $\left[\frac{1}{2}-y,\, \frac{3}{4},\, -y\right]$ | [60,168,252,360] |
| 55 | $\left[y+\frac{1}{2},\, -y,\, \frac{1}{4}\right]$ | [63,119,255,311] |
| 56 | $\left[\frac{1}{2}-y,\, -y,\, \frac{3}{4}\right]$ | [64,117,256,309] |
| 57 | $\left[y,\, \frac{3}{4},\, y+\frac{1}{2}\right]$ | [65,151,257,343] |
| 58 | $\left[y,\, y+\frac{1}{2},\, \frac{3}{4}\right]$ | [66,100,258,292] |
| 59 | $\left[-y,\, y+\frac{1}{2},\, \frac{1}{4}\right]$ | [67,101,259,293] |
| 60 | $\left[-y,\, \frac{1}{4},\, y+\frac{1}{2}\right]$ | [68,150,260,342] |
| 61 | $\left[-y,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$ | [69,112,261,304] |
| 62 | $\left[y,\, \frac{1}{4},\, \frac{1}{2}-y\right]$ | [70,155,262,347] |
| 63 | $\left[y,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$ | [71,111,263,303] |
| 64 | $\left[-y,\, \frac{3}{4},\, \frac{1}{2}-y\right]$ | [72,156,264,348] |
| 65 | $\left[\frac{1}{2}-y,\, -y,\, \frac{1}{4}\right]$ | [76,138,268,330] |
| 66 | $\left[y+\frac{1}{2},\, -y,\, \frac{3}{4}\right]$ | [77,139,269,331] |
| 67 | $\left[y+\frac{1}{2},\, \frac{3}{4},\, -y\right]$ | [78,188,270,380] |
| 68 | $\left[\frac{1}{2}-y,\, \frac{1}{4},\, -y\right]$ | [79,185,271,377] |
| 69 | $\left[\frac{1}{2}-y,\, \frac{3}{4},\, y\right]$ | [83,190,275,382] |
| 70 | $\left[y+\frac{1}{2},\, \frac{1}{4},\, y\right]$ | [84,192,276,384] |
| 71 | $\left[\frac{1}{2}-y,\, y,\, \frac{3}{4}\right]$ | [87,143,279,335] |
| 72 | $\left[y+\frac{1}{2},\, y,\, \frac{1}{4}\right]$ | [88,141,280,333] |
| 73 | $\left[-y,\, \frac{1}{4},\, \frac{1}{2}-y\right]$ | [89,175,281,367] |
| 74 | $\left[-y,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$ | [90,124,282,316] |
| 75 | $\left[y,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$ | [91,125,283,317] |
| 76 | $\left[y,\, \frac{3}{4},\, \frac{1}{2}-y\right]$ | [92,174,284,366] |
| 77 | $\left[y,\, y+\frac{1}{2},\, \frac{1}{4}\right]$ | [93,136,285,328] |
| 78 | $\left[-y,\, \frac{3}{4},\, y+\frac{1}{2}\right]$ | [94,179,286,371] |
| 79 | $\left[-y,\, y+\frac{1}{2},\, \frac{3}{4}\right]$ | [95,135,287,327] |
| 80 | $\left[y,\, \frac{1}{4},\, y+\frac{1}{2}\right]$ | [96,180,288,372] |
| 81 | $\left[\frac{3}{4},\, y,\, y+\frac{1}{2}\right]$ | [97,157,289,349] |
| 82 | $\left[\frac{1}{4},\, \frac{1}{2}-y,\, y\right]$ | [98,154,290,346] |
| 83 | $\left[\frac{1}{4},\, y+\frac{1}{2},\, -y\right]$ | [99,153,291,345] |
| 84 | $\left[\frac{3}{4},\, -y,\, \frac{1}{2}-y\right]$ | [104,158,296,350] |
| 85 | $\left[\frac{1}{4},\, y,\, \frac{1}{2}-y\right]$ | [105,147,297,339] |
| 86 | $\left[\frac{1}{4},\, -y,\, y+\frac{1}{2}\right]$ | [106,146,298,338] |
| 87 | $\left[\frac{3}{4},\, y+\frac{1}{2},\, y\right]$ | [109,145,301,337] |
| 88 | $\left[\frac{3}{4},\, \frac{1}{2}-y,\, -y\right]$ | [110,152,302,344] |
| 89 | $\left[\frac{1}{4},\, -y,\, \frac{1}{2}-y\right]$ | [121,181,313,373] |
| 90 | $\left[\frac{3}{4},\, y+\frac{1}{2},\, -y\right]$ | [122,178,314,370] |
| 91 | $\left[\frac{3}{4},\, \frac{1}{2}-y,\, y\right]$ | [123,177,315,369] |
| 92 | $\left[\frac{1}{4},\, y,\, y+\frac{1}{2}\right]$ | [128,182,320,374] |
| 93 | $\left[\frac{3}{4},\, -y,\, y+\frac{1}{2}\right]$ | [129,171,321,363] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 94 | $\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$ | [130,170,322,362] |
| 95 | $\left[\frac{1}{4}, \frac{1}{2}-y, -y\right]$ | [133,169,325,361] |
| 96 | $\left[\frac{1}{4}, y+\frac{1}{2}, y\right]$ | [134,176,326,368] |

Table 9: Wyckoff site: 96i, site symmetry: m..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, z]$ | [1,32,193,224] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$ | [2,27,194,219] |
| 3 | $\left[\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$ | [3,26,195,218] |
| 4 | $\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$ | [4,40,196,232] |
| 5 | $\left[\frac{1}{2}-z, y+\frac{1}{2}, \frac{1}{2}\right]$ | [5,39,197,231] |
| 6 | $\left[\frac{1}{2}-y, \frac{1}{2}, z+\frac{1}{2}\right]$ | [6,35,198,227] |
| 7 | $\left[y+\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$ | [7,36,199,228] |
| 8 | $[0, -y, -z]$ | [8,25,200,217] |
| 9 | $[0, y, -z]$ | [9,34,201,226] |
| 10 | $[0, -y, z]$ | [10,33,202,225] |
| 11 | $\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$ | [11,30,203,222] |
| 12 | $\left[\frac{1}{2}-y, \frac{1}{2}, \frac{1}{2}-z\right]$ | [12,31,204,223] |
| 13 | $\left[\frac{1}{2}, z+\frac{1}{2}, y+\frac{1}{2}\right]$ | [13,38,205,230] |
| 14 | $\left[\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-y\right]$ | [14,37,206,229] |
| 15 | $\left[z+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$ | [15,29,207,221] |
| 16 | $\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}\right]$ | [16,28,208,220] |
| 17 | $[z, 0, y]$ | [17,48,209,240] |
| 18 | $[y, z, 0]$ | [18,45,210,237] |
| 19 | $[-y, z, 0]$ | [19,47,211,239] |
| 20 | $[-z, 0, y]$ | [20,46,212,238] |
| 21 | $[-y, -z, 0]$ | [21,42,213,234] |
| 22 | $[z, 0, -y]$ | [22,44,214,236] |
| 23 | $[y, -z, 0]$ | [23,43,215,235] |
| 24 | $[-z, 0, -y]$ | [24,41,216,233] |
| 25 | $\left[0, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [49,80,241,272] |
| 26 | $\left[\frac{1}{2}, -z, y\right]$ | [50,75,242,267] |
| 27 | $\left[\frac{1}{2}, z, -y\right]$ | [51,74,243,266] |
| 28 | $\left[z+\frac{1}{2}, y, 0\right]$ | [52,88,244,280] |
| 29 | $\left[\frac{1}{2}-z, y, 0\right]$ | [53,87,245,279] |
| 30 | $\left[\frac{1}{2}-y, 0, z\right]$ | [54,83,246,275] |
| 31 | $\left[y+\frac{1}{2}, 0, z\right]$ | [55,84,247,276] |
| 32 | $\left[0, \frac{1}{2}-y, \frac{1}{2}-z\right]$ | [56,73,248,265] |
| 33 | $\left[0, y+\frac{1}{2}, \frac{1}{2}-z\right]$ | [57,82,249,274] |
| 34 | $\left[0, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [58,81,250,273] |
| 35 | $\left[y+\frac{1}{2}, 0, -z\right]$ | [59,78,251,270] |
| 36 | $\left[\frac{1}{2}-y, 0, -z\right]$ | [60,79,252,271] |
| 37 | $\left[\frac{1}{2}, z, y\right]$ | [61,86,253,278] |
| 38 | $\left[\frac{1}{2}, -z, -y\right]$ | [62,85,254,277] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 39 | $\left[z+\frac{1}{2},\,-y,\,0\right]$ | [63,77,255,269] |
| 40 | $\left[\frac{1}{2}-z,\,-y,\,0\right]$ | [64,76,256,268] |
| 41 | $\left[z,\,\frac{1}{2},\,y+\frac{1}{2}\right]$ | [65,96,257,288] |
| 42 | $\left[y,\,z+\frac{1}{2},\,\frac{1}{2}\right]$ | [66,93,258,285] |
| 43 | $\left[-y,\,z+\frac{1}{2},\,\frac{1}{2}\right]$ | [67,95,259,287] |
| 44 | $\left[-z,\,\frac{1}{2},\,y+\frac{1}{2}\right]$ | [68,94,260,286] |
| 45 | $\left[-y,\,\frac{1}{2}-z,\,\frac{1}{2}\right]$ | [69,90,261,282] |
| 46 | $\left[z,\,\frac{1}{2},\,\frac{1}{2}-y\right]$ | [70,92,262,284] |
| 47 | $\left[y,\,\frac{1}{2}-z,\,\frac{1}{2}\right]$ | [71,91,263,283] |
| 48 | $\left[-z,\,\frac{1}{2},\,\frac{1}{2}-y\right]$ | [72,89,264,281] |
| 49 | $\left[\frac{1}{2},\,y,\,z+\frac{1}{2}\right]$ | [97,128,289,320] |
| 50 | $\left[0,\,\frac{1}{2}-z,\,y\right]$ | [98,123,290,315] |
| 51 | $\left[0,\,z+\frac{1}{2},\,-y\right]$ | [99,122,291,314] |
| 52 | $\left[z,\,y+\frac{1}{2},\,0\right]$ | [100,136,292,328] |
| 53 | $\left[-z,\,y+\frac{1}{2},\,0\right]$ | [101,135,293,327] |
| 54 | $\left[-y,\,\frac{1}{2},\,z\right]$ | [102,131,294,323] |
| 55 | $\left[y,\,\frac{1}{2},\,z\right]$ | [103,132,295,324] |
| 56 | $\left[\frac{1}{2},\,-y,\,\frac{1}{2}-z\right]$ | [104,121,296,313] |
| 57 | $\left[\frac{1}{2},\,y,\,\frac{1}{2}-z\right]$ | [105,130,297,322] |
| 58 | $\left[\frac{1}{2},\,-y,\,z+\frac{1}{2}\right]$ | [106,129,298,321] |
| 59 | $\left[y,\,\frac{1}{2},\,-z\right]$ | [107,126,299,318] |
| 60 | $\left[-y,\,\frac{1}{2},\,-z\right]$ | [108,127,300,319] |
| 61 | $\left[0,\,z+\frac{1}{2},\,y\right]$ | [109,134,301,326] |
| 62 | $\left[0,\,\frac{1}{2}-z,\,-y\right]$ | [110,133,302,325] |
| 63 | $\left[z,\,\frac{1}{2}-y,\,0\right]$ | [111,125,303,317] |
| 64 | $\left[-z,\,\frac{1}{2}-y,\,0\right]$ | [112,124,304,316] |
| 65 | $\left[z+\frac{1}{2},\,0,\,y+\frac{1}{2}\right]$ | [113,144,305,336] |
| 66 | $\left[y+\frac{1}{2},\,z,\,\frac{1}{2}\right]$ | [114,141,306,333] |
| 67 | $\left[\frac{1}{2}-y,\,z,\,\frac{1}{2}\right]$ | [115,143,307,335] |
| 68 | $\left[\frac{1}{2}-z,\,0,\,y+\frac{1}{2}\right]$ | [116,142,308,334] |
| 69 | $\left[\frac{1}{2}-y,\,-z,\,\frac{1}{2}\right]$ | [117,138,309,330] |
| 70 | $\left[z+\frac{1}{2},\,0,\,\frac{1}{2}-y\right]$ | [118,140,310,332] |
| 71 | $\left[y+\frac{1}{2},\,-z,\,\frac{1}{2}\right]$ | [119,139,311,331] |
| 72 | $\left[\frac{1}{2}-z,\,0,\,\frac{1}{2}-y\right]$ | [120,137,312,329] |
| 73 | $\left[\frac{1}{2},\,y+\frac{1}{2},\,z\right]$ | [145,176,337,368] |
| 74 | $\left[0,\,-z,\,y+\frac{1}{2}\right]$ | [146,171,338,363] |
| 75 | $\left[0,\,z,\,\frac{1}{2}-y\right]$ | [147,170,339,362] |
| 76 | $\left[z,\,y,\,\frac{1}{2}\right]$ | [148,184,340,376] |
| 77 | $\left[-z,\,y,\,\frac{1}{2}\right]$ | [149,183,341,375] |
| 78 | $\left[-y,\,0,\,z+\frac{1}{2}\right]$ | [150,179,342,371] |
| 79 | $\left[y,\,0,\,z+\frac{1}{2}\right]$ | [151,180,343,372] |
| 80 | $\left[\frac{1}{2},\,\frac{1}{2}-y,\,-z\right]$ | [152,169,344,361] |
| 81 | $\left[\frac{1}{2},\,y+\frac{1}{2},\,-z\right]$ | [153,178,345,370] |
| 82 | $\left[\frac{1}{2},\,\frac{1}{2}-y,\,z\right]$ | [154,177,346,369] |
| 83 | $\left[y,\,0,\,\frac{1}{2}-z\right]$ | [155,174,347,366] |
| 84 | $\left[-y,\,0,\,\frac{1}{2}-z\right]$ | [156,175,348,367] |
| 85 | $\left[0,\,z,\,y+\frac{1}{2}\right]$ | [157,182,349,374] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 86 | $\left[0,\, -z,\, \frac{1}{2}-y\right]$ | [158,181,350,373] |
| 87 | $\left[z,\, -y,\, \frac{1}{2}\right]$ | [159,173,351,365] |
| 88 | $\left[-z,\, -y,\, \frac{1}{2}\right]$ | [160,172,352,364] |
| 89 | $\left[z+\frac{1}{2},\, \frac{1}{2},\, y\right]$ | [161,192,353,384] |
| 90 | $\left[y+\frac{1}{2},\, z+\frac{1}{2},\, 0\right]$ | [162,189,354,381] |
| 91 | $\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, 0\right]$ | [163,191,355,383] |
| 92 | $\left[\frac{1}{2}-z,\, \frac{1}{2},\, y\right]$ | [164,190,356,382] |
| 93 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, 0\right]$ | [165,186,357,378] |
| 94 | $\left[z+\frac{1}{2},\, \frac{1}{2},\, -y\right]$ | [166,188,358,380] |
| 95 | $\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, 0\right]$ | [167,187,359,379] |
| 96 | $\left[\frac{1}{2}-z,\, \frac{1}{2},\, -y\right]$ | [168,185,360,377] |

Table 10: Wyckoff site: 192j, site symmetry: 11'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1,193] |
| 2 | $\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$ | [2,194] |
| 3 | $\left[x+\frac{1}{2},\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$ | [3,195] |
| 4 | $\left[z+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$ | [4,196] |
| 5 | $\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$ | [5,197] |
| 6 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [6,198] |
| 7 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [7,199] |
| 8 | $[x,\, -y,\, -z]$ | [8,200] |
| 9 | $[-x,\, y,\, -z]$ | [9,201] |
| 10 | $[-x,\, -y,\, z]$ | [10,202] |
| 11 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [11,203] |
| 12 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [12,204] |
| 13 | $\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$ | [13,205] |
| 14 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$ | [14,206] |
| 15 | $\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$ | [15,207] |
| 16 | $\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$ | [16,208] |
| 17 | $[z,\, x,\, y]$ | [17,209] |
| 18 | $[y,\, z,\, x]$ | [18,210] |
| 19 | $[-y,\, z,\, -x]$ | [19,211] |
| 20 | $[-z,\, -x,\, y]$ | [20,212] |
| 21 | $[-y,\, -z,\, x]$ | [21,213] |
| 22 | $[z,\, -x,\, -y]$ | [22,214] |
| 23 | $[y,\, -z,\, -x]$ | [23,215] |
| 24 | $[-z,\, x,\, -y]$ | [24,216] |
| 25 | $[-x,\, -y,\, -z]$ | [25,217] |
| 26 | $\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$ | [26,218] |
| 27 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$ | [27,219] |
| 28 | $\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$ | [28,220] |
| 29 | $\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$ | [29,221] |
| 30 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [30,222] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 31 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [31,223] |
| 32 | $[-x,\, y,\, z]$ | [32,224] |
| 33 | $[x,\, -y,\, z]$ | [33,225] |
| 34 | $[x,\, y,\, -z]$ | [34,226] |
| 35 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [35,227] |
| 36 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [36,228] |
| 37 | $\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$ | [37,229] |
| 38 | $\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$ | [38,230] |
| 39 | $\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$ | [39,231] |
| 40 | $\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$ | [40,232] |
| 41 | $[-z,\, -x,\, -y]$ | [41,233] |
| 42 | $[-y,\, -z,\, -x]$ | [42,234] |
| 43 | $[y,\, -z,\, x]$ | [43,235] |
| 44 | $[z,\, x,\, -y]$ | [44,236] |
| 45 | $[y,\, z,\, -x]$ | [45,237] |
| 46 | $[-z,\, x,\, y]$ | [46,238] |
| 47 | $[-y,\, z,\, x]$ | [47,239] |
| 48 | $[z,\, -x,\, y]$ | [48,240] |
| 49 | $\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [49,241] |
| 50 | $\left[x+\frac{1}{2},\, -z,\, y\right]$ | [50,242] |
| 51 | $\left[x+\frac{1}{2},\, z,\, -y\right]$ | [51,243] |
| 52 | $\left[z+\frac{1}{2},\, y,\, -x\right]$ | [52,244] |
| 53 | $\left[\frac{1}{2}-z,\, y,\, x\right]$ | [53,245] |
| 54 | $\left[\frac{1}{2}-y,\, x,\, z\right]$ | [54,246] |
| 55 | $\left[y+\frac{1}{2},\, -x,\, z\right]$ | [55,247] |
| 56 | $\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [56,248] |
| 57 | $\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [57,249] |
| 58 | $\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [58,250] |
| 59 | $\left[y+\frac{1}{2},\, x,\, -z\right]$ | [59,251] |
| 60 | $\left[\frac{1}{2}-y,\, -x,\, -z\right]$ | [60,252] |
| 61 | $\left[\frac{1}{2}-x,\, z,\, y\right]$ | [61,253] |
| 62 | $\left[\frac{1}{2}-x,\, -z,\, -y\right]$ | [62,254] |
| 63 | $\left[z+\frac{1}{2},\, -y,\, x\right]$ | [63,255] |
| 64 | $\left[\frac{1}{2}-z,\, -y,\, -x\right]$ | [64,256] |
| 65 | $\left[z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$ | [65,257] |
| 66 | $\left[y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$ | [66,258] |
| 67 | $\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$ | [67,259] |
| 68 | $\left[-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$ | [68,260] |
| 69 | $\left[-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$ | [69,261] |
| 70 | $\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$ | [70,262] |
| 71 | $\left[y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$ | [71,263] |
| 72 | $\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$ | [72,264] |
| 73 | $\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [73,265] |
| 74 | $\left[\frac{1}{2}-x,\, z,\, -y\right]$ | [74,266] |
| 75 | $\left[\frac{1}{2}-x,\, -z,\, y\right]$ | [75,267] |
| 76 | $\left[\frac{1}{2}-z,\, -y,\, x\right]$ | [76,268] |
| 77 | $\left[z+\frac{1}{2},\, -y,\, -x\right]$ | [77,269] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 78 | $\left[y+\frac{1}{2}, -x, -z\right]$ | [78,270] |
| 79 | $\left[\frac{1}{2}-y, x, -z\right]$ | [79,271] |
| 80 | $\left[-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [80,272] |
| 81 | $\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [81,273] |
| 82 | $\left[x, y+\frac{1}{2}, \frac{1}{2}-z\right]$ | [82,274] |
| 83 | $\left[\frac{1}{2}-y, -x, z\right]$ | [83,275] |
| 84 | $\left[y+\frac{1}{2}, x, z\right]$ | [84,276] |
| 85 | $\left[x+\frac{1}{2}, -z, -y\right]$ | [85,277] |
| 86 | $\left[x+\frac{1}{2}, z, y\right]$ | [86,278] |
| 87 | $\left[\frac{1}{2}-z, y, -x\right]$ | [87,279] |
| 88 | $\left[z+\frac{1}{2}, y, x\right]$ | [88,280] |
| 89 | $\left[-z, \frac{1}{2}-x, \frac{1}{2}-y\right]$ | [89,281] |
| 90 | $\left[-y, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | [90,282] |
| 91 | $\left[y, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | [91,283] |
| 92 | $\left[z, x+\frac{1}{2}, \frac{1}{2}-y\right]$ | [92,284] |
| 93 | $\left[y, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | [93,285] |
| 94 | $\left[-z, x+\frac{1}{2}, y+\frac{1}{2}\right]$ | [94,286] |
| 95 | $\left[-y, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | [95,287] |
| 96 | $\left[z, \frac{1}{2}-x, y+\frac{1}{2}\right]$ | [96,288] |
| 97 | $\left[x+\frac{1}{2}, y, z+\frac{1}{2}\right]$ | [97,289] |
| 98 | $\left[x, \frac{1}{2}-z, y\right]$ | [98,290] |
| 99 | $\left[x, z+\frac{1}{2}, -y\right]$ | [99,291] |
| 100 | $\left[z, y+\frac{1}{2}, -x\right]$ | [100,292] |
| 101 | $\left[-z, y+\frac{1}{2}, x\right]$ | [101,293] |
| 102 | $\left[-y, x+\frac{1}{2}, z\right]$ | [102,294] |
| 103 | $\left[y, \frac{1}{2}-x, z\right]$ | [103,295] |
| 104 | $\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$ | [104,296] |
| 105 | $\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$ | [105,297] |
| 106 | $\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$ | [106,298] |
| 107 | $\left[y, x+\frac{1}{2}, -z\right]$ | [107,299] |
| 108 | $\left[-y, \frac{1}{2}-x, -z\right]$ | [108,300] |
| 109 | $\left[-x, z+\frac{1}{2}, y\right]$ | [109,301] |
| 110 | $\left[-x, \frac{1}{2}-z, -y\right]$ | [110,302] |
| 111 | $\left[z, \frac{1}{2}-y, x\right]$ | [111,303] |
| 112 | $\left[-z, \frac{1}{2}-y, -x\right]$ | [112,304] |
| 113 | $\left[z+\frac{1}{2}, x, y+\frac{1}{2}\right]$ | [113,305] |
| 114 | $\left[y+\frac{1}{2}, z, x+\frac{1}{2}\right]$ | [114,306] |
| 115 | $\left[\frac{1}{2}-y, z, \frac{1}{2}-x\right]$ | [115,307] |
| 116 | $\left[\frac{1}{2}-z, -x, y+\frac{1}{2}\right]$ | [116,308] |
| 117 | $\left[\frac{1}{2}-y, -z, x+\frac{1}{2}\right]$ | [117,309] |
| 118 | $\left[z+\frac{1}{2}, -x, \frac{1}{2}-y\right]$ | [118,310] |
| 119 | $\left[y+\frac{1}{2}, -z, \frac{1}{2}-x\right]$ | [119,311] |
| 120 | $\left[\frac{1}{2}-z, x, \frac{1}{2}-y\right]$ | [120,312] |
| 121 | $\left[\frac{1}{2}-x, -y, \frac{1}{2}-z\right]$ | [121,313] |
| 122 | $\left[-x, z+\frac{1}{2}, -y\right]$ | [122,314] |
| 123 | $\left[-x, \frac{1}{2}-z, y\right]$ | [123,315] |
| 124 | $\left[-z, \frac{1}{2}-y, x\right]$ | [124,316] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 125 | $\left[z,\ \frac{1}{2}-y,\ -x\right]$ | [125,317] |
| 126 | $\left[y,\ \frac{1}{2}-x,\ -z\right]$ | [126,318] |
| 127 | $\left[-y,\ x+\frac{1}{2},\ -z\right]$ | [127,319] |
| 128 | $\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$ | [128,320] |
| 129 | $\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | [129,321] |
| 130 | $\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$ | [130,322] |
| 131 | $\left[-y,\ \frac{1}{2}-x,\ z\right]$ | [131,323] |
| 132 | $\left[y,\ x+\frac{1}{2},\ z\right]$ | [132,324] |
| 133 | $\left[x,\ \frac{1}{2}-z,\ -y\right]$ | [133,325] |
| 134 | $\left[x,\ z+\frac{1}{2},\ y\right]$ | [134,326] |
| 135 | $\left[-z,\ y+\frac{1}{2},\ -x\right]$ | [135,327] |
| 136 | $\left[z,\ y+\frac{1}{2},\ x\right]$ | [136,328] |
| 137 | $\left[\frac{1}{2}-z,\ -x,\ \frac{1}{2}-y\right]$ | [137,329] |
| 138 | $\left[\frac{1}{2}-y,\ -z,\ \frac{1}{2}-x\right]$ | [138,330] |
| 139 | $\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$ | [139,331] |
| 140 | $\left[z+\frac{1}{2},\ x,\ \frac{1}{2}-y\right]$ | [140,332] |
| 141 | $\left[y+\frac{1}{2},\ z,\ \frac{1}{2}-x\right]$ | [141,333] |
| 142 | $\left[\frac{1}{2}-z,\ x,\ y+\frac{1}{2}\right]$ | [142,334] |
| 143 | $\left[\frac{1}{2}-y,\ z,\ x+\frac{1}{2}\right]$ | [143,335] |
| 144 | $\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$ | [144,336] |
| 145 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | [145,337] |
| 146 | $\left[x,\ -z,\ y+\frac{1}{2}\right]$ | [146,338] |
| 147 | $\left[x,\ z,\ \frac{1}{2}-y\right]$ | [147,339] |
| 148 | $\left[z,\ y,\ \frac{1}{2}-x\right]$ | [148,340] |
| 149 | $\left[-z,\ y,\ x+\frac{1}{2}\right]$ | [149,341] |
| 150 | $\left[-y,\ x,\ z+\frac{1}{2}\right]$ | [150,342] |
| 151 | $\left[y,\ -x,\ z+\frac{1}{2}\right]$ | [151,343] |
| 152 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | [152,344] |
| 153 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$ | [153,345] |
| 154 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | [154,346] |
| 155 | $\left[y,\ x,\ \frac{1}{2}-z\right]$ | [155,347] |
| 156 | $\left[-y,\ -x,\ \frac{1}{2}-z\right]$ | [156,348] |
| 157 | $\left[-x,\ z,\ y+\frac{1}{2}\right]$ | [157,349] |
| 158 | $\left[-x,\ -z,\ \frac{1}{2}-y\right]$ | [158,350] |
| 159 | $\left[z,\ -y,\ x+\frac{1}{2}\right]$ | [159,351] |
| 160 | $\left[-z,\ -y,\ \frac{1}{2}-x\right]$ | [160,352] |
| 161 | $\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$ | [161,353] |
| 162 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$ | [162,354] |
| 163 | $\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x\right]$ | [163,355] |
| 164 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$ | [164,356] |
| 165 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$ | [165,357] |
| 166 | $\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$ | [166,358] |
| 167 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$ | [167,359] |
| 168 | $\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y\right]$ | [168,360] |
| 169 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$ | [169,361] |
| 170 | $\left[-x,\ z,\ \frac{1}{2}-y\right]$ | [170,362] |
| 171 | $\left[-x,\ -z,\ y+\frac{1}{2}\right]$ | [171,363] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 172 | $\left[-z,\, -y,\, x+\frac{1}{2}\right]$ | [172,364] |
| 173 | $\left[z,\, -y,\, \frac{1}{2}-x\right]$ | [173,365] |
| 174 | $\left[y,\, -x,\, \frac{1}{2}-z\right]$ | [174,366] |
| 175 | $\left[-y,\, x,\, \frac{1}{2}-z\right]$ | [175,367] |
| 176 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$ | [176,368] |
| 177 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$ | [177,369] |
| 178 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$ | [178,370] |
| 179 | $\left[-y,\, -x,\, z+\frac{1}{2}\right]$ | [179,371] |
| 180 | $\left[y,\, x,\, z+\frac{1}{2}\right]$ | [180,372] |
| 181 | $\left[x,\, -z,\, \frac{1}{2}-y\right]$ | [181,373] |
| 182 | $\left[x,\, z,\, y+\frac{1}{2}\right]$ | [182,374] |
| 183 | $\left[-z,\, y,\, \frac{1}{2}-x\right]$ | [183,375] |
| 184 | $\left[z,\, y,\, x+\frac{1}{2}\right]$ | [184,376] |
| 185 | $\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, -y\right]$ | [185,377] |
| 186 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, -x\right]$ | [186,378] |
| 187 | $\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, x\right]$ | [187,379] |
| 188 | $\left[z+\frac{1}{2},\, x+\frac{1}{2},\, -y\right]$ | [188,380] |
| 189 | $\left[y+\frac{1}{2},\, z+\frac{1}{2},\, -x\right]$ | [189,381] |
| 190 | $\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$ | [190,382] |
| 191 | $\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$ | [191,383] |
| 192 | $\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, y\right]$ | [192,384] |